

REAL ESTATE PROPERTY MANAGEMENT SYSTEM

SHORT DOCUMENTATION

Proponents:

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Course and Section:

CC002.23 - BSCS-1A

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Problem Description / Background

In real estate businesses, property details, buyer information, and sales records are often difficult to track manually. This project aims to simulate a simple property management system that allows users to record listings, buyers, and sales using arrays in a console-based Java application.

The program provides an organized way to manage property transactions without the need for a database or external file.

Objectives of the Program

1. To create a simple console-based program that manages property listings, buyers, and sales.
2. To apply core programming concepts such as arrays, methods, and conditional statements.
3. To develop logical thinking and structured programming skills.
4. To provide a basic example of how property management systems operate.

Features and Functions:

- **Main Menu:** Provides a loop-based interface to navigate through options.
- **Add Property Information:** Records property ID, location, and price.
- **Add Buyer Information :** Records buyer name, contact information, and budget.
- **Record Sales:** Matches a buyer to a property and updates the property's status to "Sold."
- **View Sales:** Displays all completed sales with buyer and property details.

Program Flow / Pseudocode

1. DEFINE DATA

LIST of Properties

LIST of Buyers

LIST of Sales

2. START PROGRAM

START MainLoop

// 3. SHOW MENU

PRINT Menu:

"1. Add Property"

"2. Add Buyer"

"3. Record Sale"

"4. View Sales"

"5. Exit"

GET User_Choice

// 4. ACT ON CHOICE

SWITCH User_Choice

CASE 1: // Add Property

IF Property_List is NOT full THEN

GET Property_ID, Location, Price

ADD new Property to list

SET its IsSold_Flag to FALSE

PRINT "Property Added"

ELSE

PRINT "List is full"

END IF

CASE 2: // Add Buyer

IF Buyer_List is NOT full THEN

GET Buyer_Name, Contact, Budget

ADD new Buyer to list

PRINT "Buyer Added"

ELSE

PRINT "List is full"

END IF

CASE 3: // Record Sale

SHOW all Properties where IsSold_Flag is FALSE

ASK user to SELECT a Property

SHOW all Buyers and their Budgets

```

ASK user to SELECT a Buyer

// Core Logic
IF selected Buyer's_Budget >= selected Property's_Price
THEN
    SET selected Property's_IsSold_Flag to TRUE
    SUBTRACT Property's_Price from Buyer's_Budget
    ADD sale details to the Sales_List
    PRINT "Sale Recorded! Buyer's new budget is ___"
ELSE
    PRINT "Buyer cannot afford this property."
END IF

CASE 4: // View Sales
    IF Sales_List is empty THEN
        PRINT "No sales recorded."
    ELSE
        LOOP through each Sale in Sales_List
            PRINT Sale details (Property, Buyer, Price)
        END LOOP
    END IF

CASE 5: // Exit
    PRINT "Thank you!"
    END MainLoop

DEFAULT:
    PRINT "Invalid option. Try again."

END MainLoop

```

Screenshots of the Program Output

- A screenshot of the main menu

```
=====
# REAL ESTATE PROPERTY MANAGEMENT SYSTEM #
=====
1. Add Property
2. Add Buyer
3. Record Sale
4. View All Sales
5. Exit
Enter choice: |
```

- Adding a property

```
=====
# REAL ESTATE PROPERTY MANAGEMENT SYSTEM #
=====
1. Add Property
2. Add Buyer
3. Record Sale
4. View All Sales
5. Exit
Enter choice: 1

--- Add Property ---
Enter Property Name/ID: Tionko
Enter Location: Davao City
Enter Price: 3000000
Property added successfully!
```

- Adding a buyer

```
=====
# REAL ESTATE PROPERTY MANAGEMENT SYSTEM #
=====

1. Add Property
2. Add Buyer
3. Record Sale
4. View All Sales
5. Exit
Enter choice: 2

--- Add Buyer ---
Enter Buyer Name: Gian
Enter Contact Number: 09568278553
Enter Budget: 5000000
Buyer added successfully!
```

- Recording a sale

```
=====
# REAL ESTATE PROPERTY MANAGEMENT SYSTEM #
=====
```

1. Add Property
2. Add Buyer
3. Record Sale
4. View All Sales
5. Exit

Enter choice: 3

--- Available Properties ---

1. Tionko | Davao City | ₱3000000.0

Select property number to sell: 1

--- Buyers ---

1. Gian | Budget: ₱5000000.0

Select buyer number: 1

Sale recorded successfully!

- Viewing sales list

```
=====
# REAL ESTATE PROPERTY MANAGEMENT SYSTEM #
=====
```

1. Add Property
2. Add Buyer
3. Record Sale
4. View All Sales
5. Exit

Enter choice: 4

--- Completed Sales ---

1. Property ID: Tionko | Buyer: Gian | Price: ₱3000000.0

Conclusion and Learnings

Through this project, both of us learned how to apply structured programming principles by planning, coding, testing, debugging, and documenting a complete program. We gained experience using arrays, methods, loops, and conditionals to manage data. This project improved our teamwork, problem-solving, and understanding of how software can organize real-world information effectively.